source ..... lylepmills alt firmwares
rev 2026.10

mini elements

_travelers guide
_for zero gravity botany

3	5	7
exciters from elements	menu pages	octave steps, locked

[poly] press · [poly] hold · › then · noon = 12 o'clock · cw / ccw = clockwise / counter

elements' exciter, borrowed into rings · a menu behind the two buttons · free alt firmware for rings

installs like any rings update · long press = 3 s

orange = hold

01 buttons

rng_01 · stock gestures kept

[poly]	voices 1 · 2 · 4 · in the menu: next page
[model]	resonator · in the menu: next value
[poly] 3 s	3 voices, odd / even rhythm (as stock)
[model] 3 s	the hidden twin of this model (as stock)
tap [poly] + [model]	frequency lock on / off (02)
[poly] › [model]	3 s: **options menu** ([poly] went down first)
[model] › [poly]	3 s: **calibration** ([model] went down first)
in the menu: [poly] + [model]	saves and leaves (LEDs flicker red / green)
power on + [model]	colour-blind mode, as stock

02 frequency lock

rng_02 · tap both

tap [poly] + [model]	all LEDs flash: red = will lock · green = will unlock · it flips when you let go
locked	[frequency] knob stops tuning · v/oct, frequency cv and its attenuverter still work
locked · rings modes	[frequency] = octave switch: 7 steps, –3 to +3 octaves
locked · mini-elements	[frequency] = exciter **timbre**
remembered	the lock and its tuning survive power off

06 hold on strum

rng_06 · menu page 3

off	stock: the knobs act live
on	each strum samples structure, brightness, damping and position (and most exciter settings) · frequency stays live
with voices	only the new voice takes the new values, so every voice rings with its own settings
a strum is	a [strum] trigger, or a v/oct note change when strum is empty

09 good to know

rng_09

32 kHz inside	down from 48 kHz, as on elements: nothing above 16 kHz
no slew	the structure, brightness, damping and position cv are no longer smoothed, so fast cv bites harder

03 five modes

rng_03 · menu page 1

classic rings	stock rings, stereo
rings + waveform	mono rings on [odd] · [even] = square, impulse train or saw at the resonator pitch: tune, sync, feedback
mini-elements	elements' exciter plays the resonator inside · stereo
mini-elements + exciter	resonator on [odd] · raw exciter on [even]: envelope it, patch it back into [in]
easter egg	disastrous peace · the knob handshake is gone, the menu is the only way in

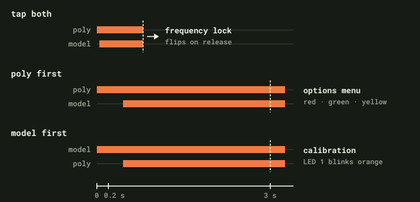
07 outputs

rng_07 · odd / even by mode

classic · mini-elements	stereo, as stock: split by partials or by voice
rings + waveform	[odd] = rings (mono) · [even] = the waveform
mini-el. + exciter	[odd] = resonator · [even] = raw exciter
in the two + modes	always patch [even]: unpatched, the hardware mixes it into [odd]

__fig.01 : three gestures

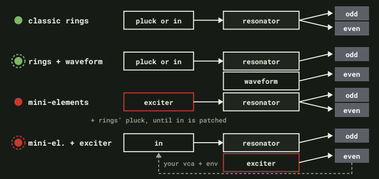
two buttons, held together



between 0.2 s and 3 s the lock waits for you to let go · past 3 s, whichever button went down first

__fig.06 : four ways out

menu page 1, drawn



04 the exciters

rng_04 · menu page 2

bow — continuous · menu page 2

timbre	bow colour, dark → bright
position	still the strike point
strum	not needed: it sounds all the time, more oscillator than resonator

blow — continuous · through elements' tube

timbre	pitch of the breath grains
position	**flow**: where the breath recording restarts
damping	also damps the tube

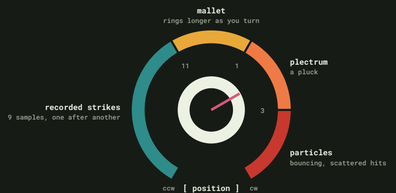
strike — needs triggers or gates on strum

timbre	how bright the hit is
position	**mallet**: recorded strikes › mallet › plectrum › particles (fig.04)
strum	each trigger or gate fires the exciter

timbre = [frequency] once locked, or the frequency cv (menu page 5) · one exciter at a time, no envelope knob · rings' own pluck still plays: a dummy cable in [in] silences it

__fig.04 : the strike dial

position = mallet



strike mode only · timbre (frequency, once locked) = how bright the hit is

08 install · calibrate

rng_08

install — free · a .wav, like any rings update

1. download rings_mini_elements.wav · lines thread
2. power on holding [poly] · play the file into [in] · [frequency] = gain
3. back to stock: the rings 1.1 .wav, same way

retune the v/oct — hold [model] first, or you land in the menu

1. unplug all cv · keyboard into [v/oct] · a cable in the [frequency] cv jack, other end loose
2. hold [model], add [poly], keep both 3 s › LED 1 blinks orange
3. send 1 v · press [poly] › send 3 v · press [poly] › done

no calibration on copies downloaded before 2022.02.27: download it again

11 who made it

rng_11 · credits

firmware	lylepmills, on émilie gillet's open rings + elements code
chord tables	jon butler · joe mcmullen

05 chord tables

rng_05 · menu page 4

classic — the stock tables

jon butler — after his plaits chords

chords	reorganised, with a few more
3–4 voices	western chords plays the first 16 intervals instead, unison to major 10th

joe mcmullen — five tables in one

easter egg	his plaits chord table, revoiced for each voice count
1 voice	15 scales (fig.05): a tuned reverb for whatever you feed into [in]
2 voices	21 chords of one key, descending in thirds: c am f dm b° g em …
3 voices	the first 16 harmonics of the overtone series
4 voices	23 just intervals after the 22 shrutis, 1/1 to 2/1

used by the easter egg and by western chords (the sympathetic strings twin) · [frequency] sets the key, [structure] picks the chord

__fig.05 : the scale dial

joe mcmullen · 1 voice



1 hungarian minor
2 lydian minor
3 melodic minor
4 harmonic minor
5 diatonic minor
6 pentatonic minor
7 minor triad
8 perfect fifths
9 major triad
10 pentatonic major
11 diatonic major
12 harmonic major
13 melodic major
14 phrygian major
15 byzantine major

minor side ccw · major side cw
set [frequency] to the key you play in

10 first patches

rng_10

bowed drone
mini-elements · bow · lock it
[frequency] now colours the bow · no strum needed, it never stops

breath pad
mini-elements · blow
[position] = flow · damping long

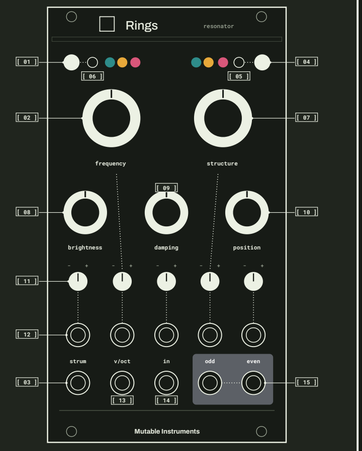
a new body every hit
strike · hold on strum on
random cv into structure, triggers into [strum]: each voice keeps its own body

shape the exciter
mini-elements + exciter
[even] › vca with an envelope › [in]

tuned reverb
[model] 3 s on sympathetic · joe table · 1 voice
any voice into [in] · [frequency] = the key

__fig.03 : front panel

what each part does here



[01] __poly : voices · menu page
[02] __frequency : timbre or octaves, once locked
[03] __strum : strike needs it
[04] __model : resonator · menu value
[05] __model LED : menu value
[06] __poly LED : menu page

[07] __structure : body · chord · scale
[08] __brightness : material
[09] __damping : decay · tube damping
[10] __position : strike point · flow · mallet
[11] __atts : one per knob
[12] __cv ins : frequency cv can drive timbre
[13] __v/oct : pitch · strums
[14] __in : dummy cable = no pluck
[15] __odd / even : see 07

__fig.02 : the options menu

left LED = page · right LED = value

mode page 1	classic rings · rings + waveform · mini-elements · mini-elements + exciter · easter egg
waveform / exciter page 2	square · bow · impulse train · blow · saw · strike
hold on strum page 3	off · on
chord table page 4	classic · jon butler · joe mcmullen
frequency cv page 5	pitch, as stock · exciter timbre

[poly] › [model] 3 s = in · [poly] = next page · [model] = next value
[poly] + [model] = save + out

◉ = blinking · in the menu the poly LED flashes every 10 s as a reminder